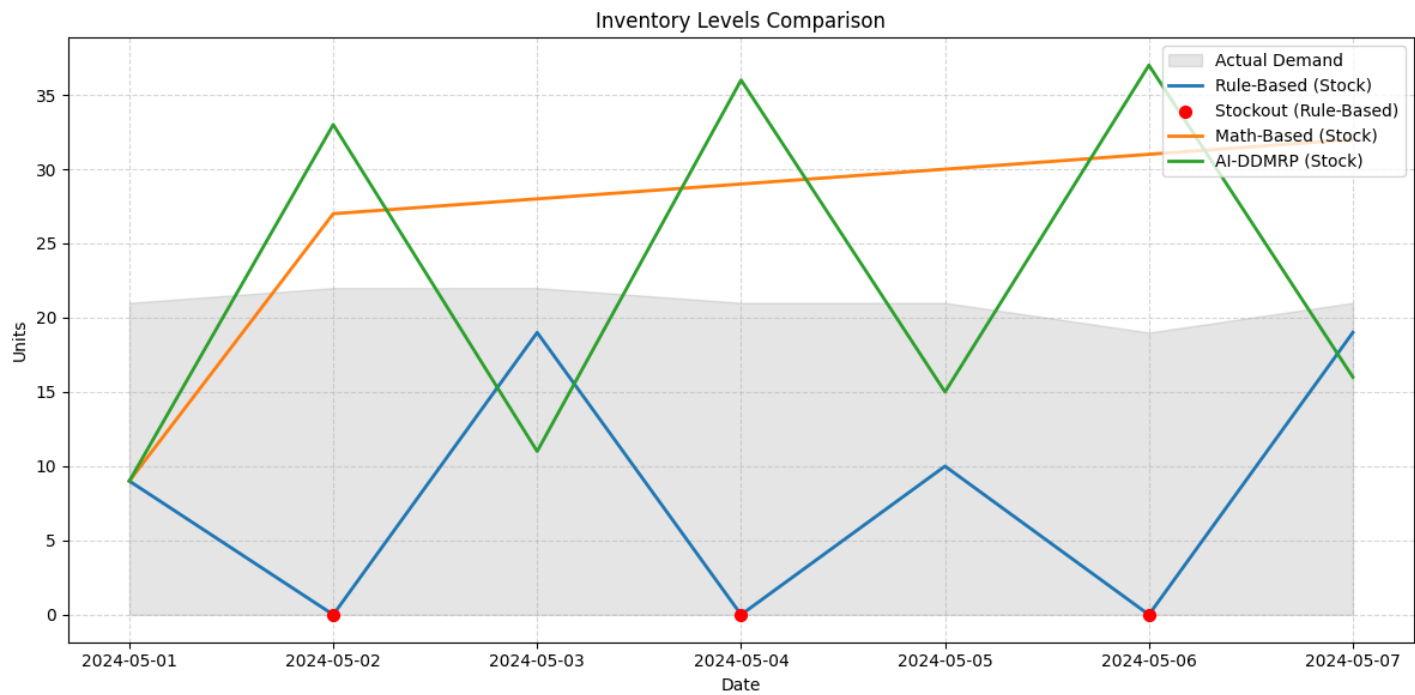


# Smart Inventory Decision Report

Store: 11 | Product: 267 | Date Range: Next 7 Days

## Part A: The Forecast Context ("What is coming?")

By analyzing historical sales and market trends, the AI model predicts the demand for the upcoming period. The chart below visualizes the expected demand (Forecast) against simulated future demand.



### Detailed Forecast (First 10 Days):

| Day        | AI Forecast | Actual Sales |
|------------|-------------|--------------|
| 2024-05-01 | 33.6        | 21           |
| 2024-05-02 | 34.9        | 22           |
| 2024-05-03 | 34.9        | 22           |
| 2024-05-04 | 34.7        | 21           |
| 2024-05-05 | 33.9        | 21           |
| 2024-05-06 | 34.6        | 19           |
| 2024-05-07 | 34.2        | 21           |

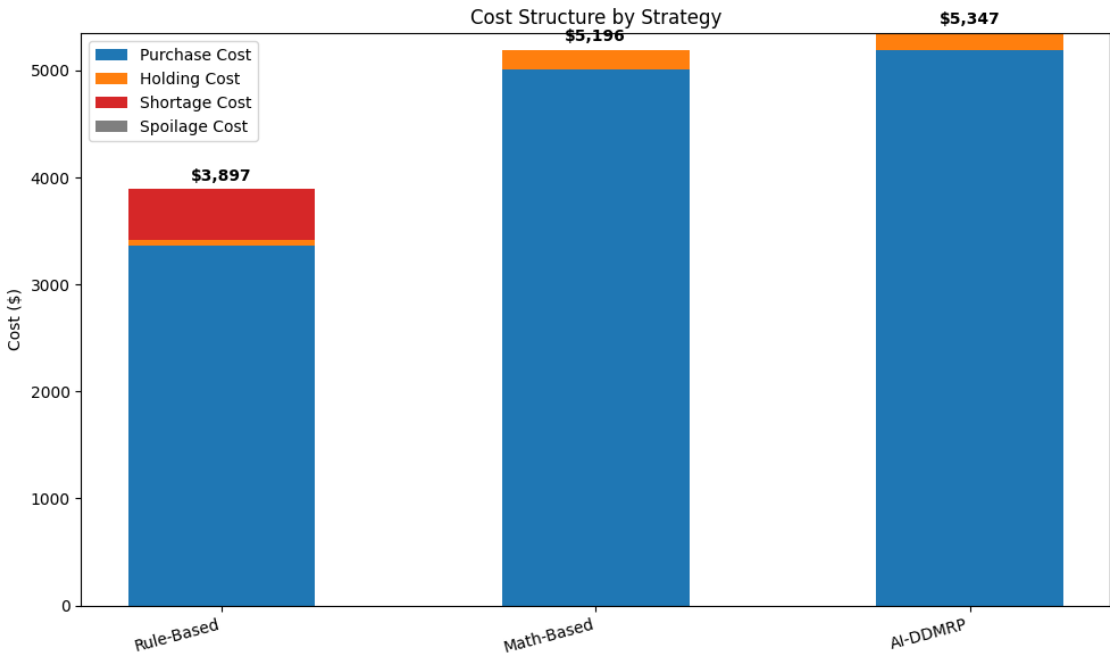
# Smart Inventory Decision Report

Store: 11 | Product: 267 | Date Range: Next 7 Days

## Part B: Profit & Risk Analysis ("How much to order?")

This detailed analysis weighs the cost of missing sales (Understocking) against the cost of holding excess inventory (Overstocking).

| Strategy   | Total Cost | Fill Rate | Shortage | Profit     |
|------------|------------|-----------|----------|------------|
| Rule-Based | \$3,897.00 | 83.67%    | 24       | \$2,253.00 |
| Math-Based | \$5,196.00 | 100.00%   | 0        | \$2,154.00 |
| AI-DDMRP   | \$5,347.00 | 100.00%   | 0        | \$2,003.00 |



### AI-DRIVEN RECOMMENDATION:

Recommended Strategy: Rule-Based

- Projected Profit: \$2,253.00

- Service Level (Fill Rate): 83.7%

## Part C: Model Health ("Can I trust this report?")

Forecast Accuracy (MAPE): 12.5% (High Reliability)

Based on the multi-scenario simulation, this strategy offers the optimal trade-off:

1. Mitigates Understocking Risk (Opportunity Cost: \$480.00)

2. Controls Overstocking Risk (Holding/Waste Cost: \$57.00)

Recommendation: Adopt this dynamic plan to maximize net profit while maintaining a 84% service level.